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A clutch of design features ensures that OKAYA SMF / VRLA batteries perform predictably and reliably every time.

- High Power Density
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- Higher Reliability & Quality

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Q1. How do the concepts of fuzzy logic and neural network differ?

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A1. Fuzzy sets. A classical set, as we know, is a well-defined collection of distinct objects called 'elements' or 'members' of the set. So we have two categories of objects; members of the set, as they are within the boundaries defined by the set, and non-members of the set, as they do not share the properties that define the set, and therefore are outside the boundaries of the set. In other words, an object is either 'in' or 'out' of the set.

We have another kind of sets, known as fuzzy sets, whose elements have degrees of membership. In the classical set, the membership of elements in a set is assessed in binary terms according to a bivalent condition—an element either belongs or does not belong to the set. By contrast, the fuzzy set theory permits gradual assessment of the membership of elements in a set; this is described with the help of a membership function valued in the real unit interval $[0, 1]$. Classical sets are special cases of fuzzy sets; if the membership functions only take values 0 or 1. The fuzzy set theory can be used in a wide range of domains in which the information is incomplete or imprecise, such as bioinformatics.

Neural networks. These emulate the behaviour of biological neural systems in digital software or hardware. They can learn complex relationships among data automatically. This feature makes the technique very useful in modeling processes.

In digital computing, data processing applications use programmed computation, which requires previous development of a mathematical or logical algorithm to solve the problem at hand. This approach can be used only when the processing to be made can be precisely described in a known rule set.

Neural networks, however, do not require previous development of algorithms or rule sets to analyse data. A

neural network is able to identify the required relationships from real data, avoiding the previous development of any model.

Difference between the programmed computation and neural networks can be best understood when we compare the operation of computers with humans. For example, a computer can perform mathematical operations more quickly and precisely than a human. Also, there is the element of consistency. However, humans can recognise faces and complex images in a more precise, efficient and faster manner than the best of the computers.

One of the reasons for this performance difference can be attributed to the distinct organisation forms of computers and biological neural systems. A computer with the help of a processor executes instructions delivered by a programmer one by one. Biological neural systems consist of billions of nervous cells (neurons) with a high degree of interconnection. Neurons can perform simple calculations without the need to be previously programmed. Artificial neural networks are made up of interconnecting artificial neurons (programming constructs that mimic the properties of biological neurons).

Applications particularly suited for neural networks are those in which mathematical modeling is difficult or impossible. Some examples are signal analysis and processing, process control, robotics, data classification, data smoothing, pattern recognition, image analysis, speech analysis, medical diagnostics and stock market forecasting.

Similarities and dissimilarities. Fuzzy logic and neural networks are dynamic systems. These estimate function from the sample data itself and therefore do not resort to mathematical modeling. Both of them process the inexact information inexactly.

The main dissimilarity between fuzzy logic and neural network is the technique employed for making the rules. The former uses heuristic knowledge using sample data to form rules, whereas the latter forms rules based